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REVIEW ARTICLE

A systematic study on the role of SentiWordNet in opinion mining

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Abstract Sentiment lexicons (SL) (aka lexical resources) are the repositories of one or several dictionaries that consist of known and precompiled sentiment terms. These lexicons play an important role in performing several different opinion mining tasks. The efficacy of the lexicon-based approaches in performing opinion mining (OM) tasks solely depends on selecting an appropriate opinion lexicon to analyze the text. Therefore, one has to explore the available sentiment lexicons and then select the most suitable resource. Among available resources, SentiWordNet (SWN) is the most widely used lexicon to perform tasks related to opinion mining. In SWN, each *synset* of WordNet is being assigned the three sentiment numerical scores; positive, negative and objective that are calculated using by a set of classifiers. In this paper, a detailed and comprehensive review of the work related to opinion mining using SentiWordNet is provided in a very distinctive way. This survey will be useful for the researchers contributing to the field of opinion mining. Following features make our contribution worthwhile and unique among the reviews of similar kind: (i) our review classifies the existing literature with respect to opinion mining tasks and subtasks (ii) it covers a very different outlook of the opinion mining field by providing in-depth discussions of the existing works at different granularity levels (word, sentences, document, aspect, clause, and concept levels) (iii) this state-of-art review covers each article in the following dimensions: the designated task performed, granularity level of the task completed, results obtained, and feature dimensions, and (iv) lastly it concludes the summary of the related articles according to the granularity levels, publishing years, related tasks (or subtasks), and types of classifiers used. In the end, major challenges and tasks related to lexicon-based approaches towards opinion mining are also discussed.

Keywords opinion mining, lexical based resources, SentiWordNet, opinion strength

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1 Introduction

Opinion mining (OM) is the general task of extracting opinions from the text documents [1–3]. An opinion is constituent of two basic components: an object and the sentiment about the object. For example, in the statement "the picture quality of my camera is amazing"; here, "camera" is the object and sentiment about the picture quality is "amazing". OM is also referred to as sentiment analysis, subjectivity analysis or sentiment mining [3–9] so we will use these terms interchangeably. The field of OM is different from the conventional information retrieval (IR) process in a way that IR emphasizes on the retrieval of relevant information matching some query. On the other hand, the purpose of OM is the process of extracting the opinion of the public about a specific product using natural language processing algorithms and text analysis tools [3, 5, 7, 9]. For example, if someone states "it is a nice weather today"; here the word "nice" is describing the situation of the weather. In other words, the user is giving her personal opinion about the weather. On the contrary, the factual statement e.g., "Two and two make four" will be valid for all people and giving a piece of information about some facts. The research problems related to OM got huge awareness since 2005 [2, 5, 7, 9–16] (Fig. 1). The reasons behind the focusing of researchers towards the OM are (i) diversity of applications in almost every domain, (ii) the nature of problems that had never been studied before and (iii) availability of big data repositories such as the online social media [12, 13]. Since then, OM is widely applied to reviews, blogs, social and print media and for a variety of applications ranging from marketing to customer services. Some other task of OM is classifying the polarity of expressed opinion in the document.

The approaches to solving OM problems are divided into three main categories. (1) Classification approaches (2) Natural Language Processing (NLP) approaches and (3) Lexicon-based approaches [5, 16–18]. In classification approaches, a machine learning classifier is used to build a training model based on the

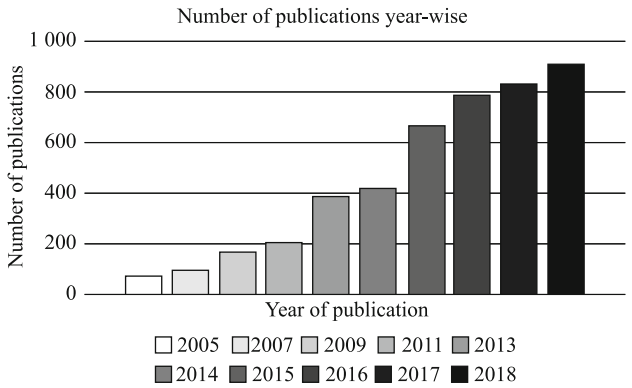

Fig. 1 Trends in number of publications in opinion mining (figure retrieved from web of science year-wise record([5, 9, 17])

already annotated data. Then by using this model the classes of the documents in the test data are estimated. In NLP-based based approaches, the information about Parts of Speech (PoS) and syntactic relationship among these PoS in each sentence of the document is estimated. While in lexicon-based approaches, the use of domain-independent lexical resources (such as SentiWordNet [7, 19], WordNet [20–22], General Inquirer [23, 24] play vital role. Furthermore, in lexical-based approaches, the systems generally make use of pre-developed glossaries containing the terms associated with the polarity score. A semantic lexical resource is defined as a database containing positive and negative words, each tagged with some calculated prior polarity score [10, 19, 25]. Among the available lexicons, SentiWordNet (SWN) is most widely used because of its converge of all the three labels namely Objective, Subjective and Negative assigned to each *synset*. Furthermore, each *synset* is also labeled using numerical scores [10, 19, 25] indicating that the sentiment is now tied to the meaning of the word rather than the complete word itself. Furthermore, *synset* is also regarded as positive and negative at the same time. These salient features distinguish SWN from other lexical resources. This review covers major contributions of SWN in the field of OM.

This paper is organized as follows. Section 2 describes the basic steps of OM with an explanation of each phase. In Section 3 the brief introduction of SWN is presented followed by the visualization of a *synset* and evaluation of SWN. Section 4 gives detailed discussion on the existing notable works of OM from word to concept level using SWN. In Section 5 detail discussion, open challenges and issues related to OM are highlighted. The Section 6 concludes the review and valuable remarks on future work.

2 Basic steps in opinion mining

Opinion mining (OM) task consists of multiple steps [18, 26–29]. Since the different types of text originate from multiple resources in varied formats, therefore, various steps are required to perform efficient OM. Data acquisition, text preprocessing, data transformation and text pre-processing are the most common steps in OM [18, 27, 30]. Each step is described in detail in the following sections.

2.1 Data acquisition

Data (or corpus) acquisition is considered as the first step of OM. In this step, the corpus is obtained from which the opinions are mined. This phase also depends on the type of technique used to build the corpus.

In data acquisition, data is obtained through approaches: (i) The first approach is by using the website's application program interface (API) and (ii) the second is through the use of web crawlers that extract the required data from different websites. Both approaches come with some inherent advantages and disadvantages so there is a compromise between using either. The work of [18] provides a brief comparison between both.

Similarly, other social media networks such as Facebook and Sina-Weibo have introduced Facebook Graph API and Tencent API respectively to extract the information from these media. Similarly, Twitter [31] provides two API's namely REST and Twitter4J to extract static and continuously streaming tweet data. The work reported in [31–35] exploited these API's in detail. Furthermore, Google Scholar is also one of the most widely used search engine. The readers are advised to consult [36] for a review on comparing PubMed, Scopus, the web of science and Google scholar.

2.2 Text preprocessing

To start an organized and systematic analysis of corpus data collected from various sources, pre-processing steps are applied to prepare the data. Some of the popular pre-processing steps are,

- *Tokenization* — to remove punctuation marks and stop words in order to break a sentence into its component words.
- *Lemmatization* — to bring a word into its root form, while ignoring other PoS of the word.
- *Stop word removal*, to remove the frequently occurring terms (or words) in corpus e.g., pronouns and articles, etc. [27, 37],
- *Parts of speech (PoS) tagging* — assigns parts of speech to each token
- *Features extraction and their representation*.

Some of the publicly available different tools for pre-processing are listed in Table 1.

2.3 Data transformation

In this phase, characteristic features of the corpus data are extracted according to the requirement of the opinion mining task. There are various models for data transformation to perform OM tasks e.g., text clustering, summarization or categorization. Bag of words, Vector Space Model (VSM), *tf-idf* model, similarity measures etc. The comprehensive detail of these models is in [42].

2.4 Data analysis

Data analysis is the task to perform computational analysis of transformed text data. The text analytics apply a set of efficient

Table 1 List of text preprocessing tools

Name of tool	Purpose	Ref.
Stanford PoS Tagger	Tagging PoS	[38]
Lancaster stemmer	Stemmer	[39]
PoS tagger	Tagging Twitter PoS	[40]
Snowball	English PoS stemmer	[41]

models and algorithms that extracts the contextual properties of the data to perform various tasks. Some of these tasks are sentiment analysis, opinion mining, topic modeling, author profiling [18].

3 SentiWordNet: a general purpose sentiment lexicon

SentiWordNet (SWN) is a global (or general) lexical resource [7, 19] developed in 2006. In SWN, each synset (denoted by *sn*) of WordNet is associated with three numerical scores, namely *Obj(sn)*, *Sub(sn)*, and *Neg(sn)*. These synsets refer to *Objective*, *Subjective* and *Negative* scores. The range of theses scores lies in the closed interval [0 1] with their cumulative sum of 1.0 for each synset. It is noteworthy that each synset may have a non-zero score for all the three categories representing the fact that corresponding terms have each of the three opinion related properties only to a certain degree. The process of assigning the scores is performed by eight ternary classifiers [7, 19] applied on WordNet synsets. These classifiers are characterized by different classification behavior but with similar accuracy levels. This process helps in determining both the semantic orientation and semantic strength of synsets rather than Subjectivity-Objectivity tagging. Finally, the scores are obtained by precisely analyzing the glosses of the synsets. Due to wide coverage of SentiWordNet, all the synsets are tagged from each of the three labels **Objective**, **Positive**, **Negative**. Furthermore, numerical scores are also assigned to these synsets to represent their semantic orientations [5, 7, 17, 25]. Currently, SWN is licensed to more than 300 research groups and has been used in many projects [7, 19]. These are the reasons we selected this lexical resource that is explicitly developed for performing OM tasks. A sample template of SWN is depicted in Fig. 2.

Table 2 depicts some statistics about the distribution of different parts of speech scores in SWN. Currently the developers of SWN are working on labeling the corpus consisting of 10,000 WordNet synsets tagged by five different evaluators. One of the phenomenal facts of SWN is that the *synsets* are expected to have some degree of opinion oriented properties. However, with the decrease in the objectivity score we get a strong indication that either the given synset is positive, or Negative, or a combination of them. The ratio of the synsets that were found involved in showing decreasing behavior rapidly is from 10.45% for Objective <= 0.5, to 0.56% for Objective <= 0.125.

Following subsections include discussion on visualization of synset and evaluation of SWN.

Table 2 (a)–(c) Scores of each of the parts of speech in SWN and (d) average obtained for each part of speech [7]

(a) Distribution of scores of adjectives			
Score	Positive/%	Negative/%	Objective/%
0.0	65.77	62.81	0.08
0.125	12.12	7.32	2.14
0.25	2.37	2.42	0.87
0.375	1.25	1.54	1.84
0.5	0.62	1.35	2.32
0.625	2.94	5.53	7.65
(b) Distribution of scores of names in SWN			
Score	Positive/%	Negative/%	Objective/%
0.0	90.80	89.25	0.00
0.125	4.53	3.93	0.23
0.25	2.37	2.42	0.87
0.375	1.25	1.54	1.84
0.5	0.862	1.35	3.32
0.625	0.24	0.91	2.57
(c) Distribution of scores of verbs in SWN			
Score	Positive/%	Negative/%	Objective/%
0.0	89.80	87.93	0.00
0.125	4.43	4.94	0.21
0.24	2.66	2.95	0.64
0.375	1.55	1.81	1.35
0.5	0.84	1.24	2.67
0.625	0.36	0.63	3.40
(d) Distribution of scores of all parts of speech in SWN			
Score	Positive/%	Negative/%	Objective/%
0.0	85.18	84.85	0.02
0.125	5.79	4.77	0.54
0.25	3.56	3.58	1.97
0.375	2.28	2.19	3.72
0.5	1.85	3.07	4.20
0.5	0.84	1.24	2.67
0.625	0.87	1.64	3.83

3.1 Graphical model of SentiWordNet

As discussed earlier that for a given synset *sn* the scores (Objective(s) + Positive(s) + Negative(s)) sum to 1 therefore it is comfy to visualize the synset *sn* scores by using a triangular format. The vertices of the triangle may represent the highest score of 1.0 for either tag and 0.0 for the other two. The center of the triangle has 0.33 for all the three tags (Objective, Positive, Negative). We can make use of values at PN-polarity and SO-polarity axes to interpret (i) Subjectivity and Objectivity (ii) Positive and Negative strength/scores for given synset [7, 19]. Figure 3 shows the graphical model designed by [7] in order to display the scores of the *synset*. This model is also used in a web-based graphical interface to explore each term in the SentiWordNet, and can be freely accessed online.

3.2 Evaluation of SentiWordNet

To evaluate the accuracy of SentiWordNet 3.0, a model is proposed in [7] that performs a comparative analysis of a small,

#	POSID	PosScore	NegScore	SynsetTerms
a	00106456	0	0.625	stingy#2 scrimpy#1 meagre#1 meagerly#1
a	00106821	0	0.5	spare#5 scanty#1 bare#2 quantity;
a	00107017	0.125	0.625	exiguous#1 extremely scanty;
a	00107128	0	0.5	hand-to-mouth#1
a	00107250	0	0.375	hardscrabble#1
a	00107384	0	0	paltry#2 miserable#6 measly#1
a	00107648	0.5	0	anabolic#2

Fig. 2 Sample template of SentiWordNet 3.0. First vector showing parts of Speech (PoS) of the synset, the second vector representing position offset in WordNet, third and fourth vectors showing positive and negative score of the synset respectively and the last one giving detail about the entries of the synset [7, 19, 43]

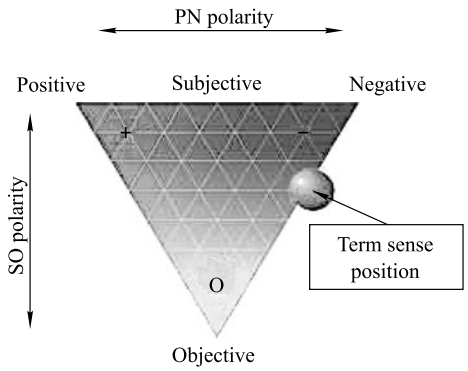

Fig. 3 Graphical model adopted by SentiWordNet to represent the score of a term sense [7, 19]

manually annotated subset of WordNet with the same set of synsets from SentiWordNet. The model proposed tagging procedure that is next to impossible since it needs full manual tagging of synsets of WordNet. Therefore, the lack of such manually tagged terms is the only reason to introduce a model to perform automatic labeling. Therefore, two approaches are proposed in [19] to evaluate the quality of SentiWordNet and thus be considered only approximately indicative of the accuracy of SentiWordNet labeling.

The main issue in using SentiWordNet for OM is to find an optimal way of combining the terms (or words) scores to compute an overall score for the whole document. The experiments of [43] concluded that the SentiWordNet scores may solve the task of detecting positivity-negativity-polarity better than other sentiment lexicons. The notable works [2, 17, 43–56] concluded that SWN-based methods for OM are robust. It is also observed that SWNN drastically improved results of sentiment analysis at any granularity level in comparison with a conventional simple baseline approach.

4 Review methodology

The current survey is undertaken in six general encompassing dimensions namely, polarity estimation, cross-domain OM, lexical creation, aspect extraction, cross, and multilingual OM, and various applications of OM. Our purpose is to assess in detail the use of SWN in the tasks mentioned above. In general, we categorized the tasks discussed above based on different levels using SWN as a lexical resource as given below:

- *Word level processing*
- *Sentence level processing*
- *Document level processing*
- *Aspect and Clause level processing*
- *Concept level processing*

4.1 Word level processing

It is noteworthy that the efficiency of any approach to classifying the polarity of the sentiment is based on the performance of the model used for said purpose. Therefore, an effective combination of the classification methods can eventually help in predicting the semantic orientation of the terms in a sentence.

The following three analysis [5, 16, 44, 58] tasks come under the umbrella of OM at word level: (i) to determine the subjectivity/objectivity of the words in the document (ii) to estimate the polarity of words i.e. the word is positive/negative in its subjectivity, and (iii) to determine the strength of the polarity. In this section, we cover the approaches focusing on any (or all) of these tasks. Sentiment analysis of microblogs like Twitter [31] faces several challenges due to some inherent problems, i.e., the limited size of posts, informal languages containing slang, etc. These slangs may have expressions like *gr8* for *great*, *FTW* for *for the win* and *LOL* for *laughing out loud* etc. that are considered as nonstandard terms. To resolve these issues, the approach proposed in [21] carefully selects the features of the tweet posts by using a machine learning technique. The authors extended these feature vectors by adding the concept of similar adjective and verb groups from tweets using SWN. After evaluating 9646 tweets, Naïve Bayes (NB) model outperformed SVM (Support Vector Machines). However, the average accuracy obtained by SVM is slightly higher than that of the NB model. Tables 3 and 4 present the results for each kind of feature vectors using SVM and NB respectively. Similar task on tweets at word level is performed in [49] using model based on multi-lexical resources such as SentiWordNet [7, 19], WordNet-Affect [8, 22], Multi-Perspective Question Answering (MPQA) Subjectivity Lexicon [59] and SenticNet [16]. The results showed that MPQA and SentiWordNet emerged as better performing lexical resources than others, see Fig. 4.

In [60] polarity estimation of the tweets is calculated by implementing a random walk algorithm [61] on WordNet graph. In this approach, the nodes of WordNet are considered as synsets and edges depict the different semantic orientations among the synsets. These semantics are analyzed using SWN to find the similarity index between different synsets. The final polarity score is computed as,

$$p = \frac{r.s}{|t|}. \tag{1}$$

Here, the notations p - final score, s - vector polarity scores from SWN, r - vector of weighted synsets computed by random walk algorithm of tweets text over WordNet, and t - a set of concepts (or features) obtained from the tweets. The key purpose is to "expand" the set of concepts with additional ones that are closer to the WordNet graph. This corresponds to synset nodes that are activated during the random walk steps. Therefore the

Table 3 Results of feature vectors using support vector machines (SVM) classifier [21]

Feature Vector		Uni-gram	adj	DBpedia	Verb groups syntactic senti-features
f-measure	**Positive**	**0.603**	0.619	0.622	**0.637**
	Negative	0.443	**0.436**	0.417	0.44
	Neutral	0.683	0.685	**0.691**	0.689
	Avg. neg.+pos.	0.523	0.527	0.52	**0.538**

Table 4 Results of feature vectors using Naïve Bayes [21] approach

Feature Vector		Uni-gram	adj	DBpedia	Verb groups syntactic senti-features
f-measure	**Positive**	0.514	**0.563**	0.562	0.540
	Negative	0.397	0.422	**0.427**	0.424
	Neutral	0.608	**0.652**	0.648	0.636
	Avg neg+pos	0.456	0.493	0.495	**0.482**

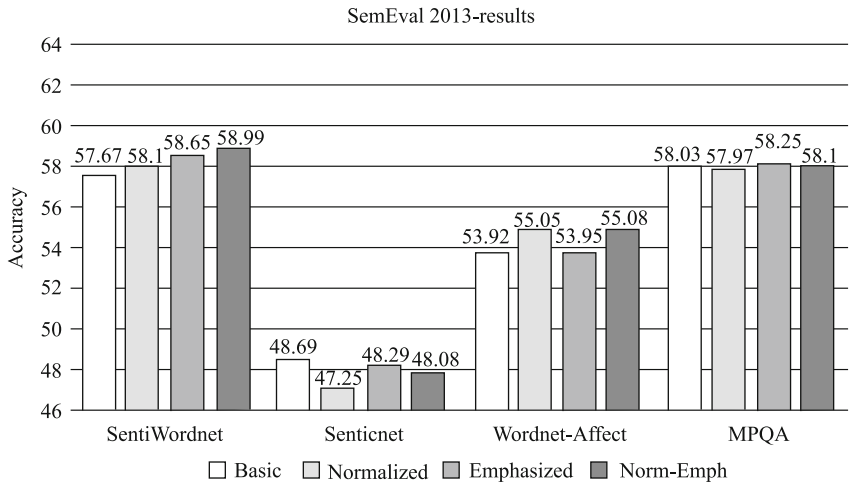

Fig. 4 Results of the comparison of experiments using four different lexicons on SemEval-2013 data [49]

terms like *dog* and *bite* (both are labeled as *neutral* in SWN) appearing in the same tweet could be extended by using a more emotive term like *hurt* holding a negative score of 0.75 in SWN. For evaluation purposes, 85, 423 tweets were analyzed without removing stop words and achieved a classification accuracy of 64% by analyzing the antonym relationship among the terms within the tweets. In order to perform semantic classification, the approaches used in [62, 63] considered only *nouns*, *adjectives*, *adverbs* and *verbs* in the given text. SWN is used to compute the overall polarity by considering all the senses of each PoS mentioned above. For example, the term *deep* has 15 adjective, three nouns and three verb senses [7, 19] in SWN, therefore, all these senses are considered to calculate the average polarity of the tweet. According to the scenario discussed in [63], the term *deep* with an adjective tag is likely to obtain the positive score because the sum of its positive scores (2.87) is much higher than the sum of its negative scores (1.25) in all of its adjective senses in SWN. The same calculation holds for its adverb PoS tag also. However, there is a different situation for the same term *deep* with a noun PoS tag. It receives a "0" label because the overall polarity for its noun senses is zero in SWN. This calculation is then reused for the evaluation purpose of small data sets from 6 different domains using the SVM classifier. The results showed that sense-level polarity information is useful in producing an efficient and productive set of seed words that may be used in improving sentiment classification.

Table 5 summarizes the distribution of articles discussed in this article, based on OM performed at different granular levels.

Sentiment analysis of opinions from word-of-mouth (WoM) enables customers to develop strategies such as buying, diffusion, switching, etc. SWN is one of the lexicons that is increasingly utilized to extract WoM sentiments. Opinion mining in e-WoM has become an attractive field among researchers. In [64, 65], SWN is used to assess the objective words by extracting the very first sense of the term in SWN. The results showed that the WoM sentiment classification can be improved by considering the objective words only.

An improvement in SWN is observed in the work of [65] in which lexicon SentiFul is developed by utilizing the WordNet-Affect database. Sentiful is a database that contains a total of 2440 entries based on nine different direct and indirect emotions. Further extension of the lexicon SentiFul was also completed by using SWN. New lexemes are generated by carefully examining bases and affixes of opinion words. In this work, a two-phase process is performed for evaluation of the lexicon: (i) First, about 200 terms were annotated using different methods and then (ii) General Inquirer (GI) based evaluation was performed and achieved accuracy of 94.0% and 86.0% for SentiFul core and SentiFul respectively. The authors described the approaches for deriving and extracting the new sentiment-carrying terms based on their synonym, antonym, hyponym relations, and derivation.

A novel approach is proposed in [66] to create an automatic sentiment sensitive thesaurus for cross-domain sentiment classification. In this approach, the efficiency of the binary classifier is increased by using the already existed thesaurus by expanding feature vectors for both training and testing purposes. Finally, a sentiment score is calculated using the point-wise mutual information (PMI) of the feature vectors and sentiment element. Experiments were performed on a dataset [67] and achieved an average accuracy of 80.9%. Similarly in [68] the authors proposed a novel method for similarity measures

Table 5 Distribution of articles on the basis of OM performed at different granular levels

Level of analysis	References	Number of articles
Word level	[21, 49, 60, 62–70]	13
Sentence level	[20, 48, 52, 55, 56, 76, 78, 80, 82, 85, 86, 92–94, 99] [77, 88, 90, 101]	19
Document level	[27, 45, 67, 92, 103–105, 108, 110, 114, 119, 125]	12
Aspect & clause level	[28, 90, 92, 112, 113, 126, 127, 129, 131, 135]	10
Concept level	[16, 133]	2
Application of OM	[16, 44, 84, 93, 93, 116–121, 132]	13

Table 6 Comparison of proposed approach using NB and SVM with some notable existing work [69]

Reference	Classifier used	Accuracy/%	t-test p-value
[3]	SVM	82.89	< 0.0001
	NB	81.00	< 0.0001
[68]	SVM	–	–
	NB	86.000	< 0.0001
[73]	SVM	88.94	0.005993
	NB	86.23	0.008053
[74]	SVM	87.15	< 0.0001
	NB	81.50	< 0.0001
[75]	SVM	81.15	< 0.0001
	NB	75.5	< 0.0001
Proposed [69]	SVM	92.35	–
	NB	82.20	

among the features extracted from opinions on electronic gadgets. For evaluation purposes, a related corpus was developed. The results indicate that using SWN in a strict sense obtains good quality features but with only a small set of features. The authors concluded that the efficient use of SWN helps in grasping more useful features.

The manual tagging of the data of blogospheres, news sites, social media networks, e-commerce sites, etc., is practically near to impossible due to its huge size. Data unavailability, data sparsity, domain independence, and inadequate performance are some of the major issues in the automation of this huge data. To manage this huge data, there is a need for an automatic approach. In [69], the authors proposed a corpus and context-based knowledge algorithm for automatic feature selection for sentiment analysis. The term scores in SWN are used for calculating statistical parameters for each semantic class represented in the one-dimensional data structure. The proposed feature selection algorithm works in two phases: first, the extracted features are grouped on the similarity basis of semantic and statistical measures. In the second phase, these features are then ranked by eliminating the low-ranked feature group. It was concluded that the diversity within the cluster of features tends to reduce the size of the feature vector. This conclusion significantly improved the performance by reducing the time complexity of the overall process. The results showed that SVM outperformed other known machine learning algorithms, as shown in Table 6.

Similar work is reported in [70] in which a semi-supervised approach for sentiment analysis is proposed. The sentiment scores defined in SWN are revised using mathematical models of cosine similarity and information gain. After performing several experiments on various datasets, it was concluded that *nouns* and *adjectives* play a better role among other parts of speech. The proposed model outperformed other related techniques and achieved the highest accuracy of 87.71% as shown in Table 7.

4.2 Sentence level processing

Social networks are one of the most indicative maneuvers of Web 2.0 that invite millions of users to publish and share information through social sites like Twitter [31], Facebook, LinkedIn, Google+, etc. These sites instigated the NLP community not only to analyze the posts on these social sites but also to explore the tasks and issues related to opinion mining. Most of the OM tasks are performed on social network data. Two major tasks are involved while performing opinion mining at sentence level: (a) identification of subjective/objective data in the sentence and (b) polarity estimation of the whole sentence. In this section, we will discuss major contributions related to these tasks in detail.

In [76] the authors determined the polarity of the textual segments within multilingual framework. Each sentiment-carrying sentence in the documents is searched for adjectives, verbs or noun in order to perform sentiment classification. With the help of SWN scores, positivity and negativity of these PoS were interpreted with the formula given below:

$$\begin{aligned} score(pos)A &= \frac{1}{n}\sum_{i=1}^{n} Score(pos\ i), \\ score(neg)A &= \frac{1}{n}\sum_{i=1}^{n} Score(neg\ i), \\ score(obj)A &= \frac{1}{n}\sum_{i=1}^{n} Score(obj\ i), \end{aligned} \tag{2}$$

Table 7 Results (%) for sentiment classification based on Senti-Cosine [70]

	Acc	Prec	Rec	FM	Acc	Prec	Rec	FM
	Feature presence				***Feature frequency***			
LMR	85.00	82.87	88.24	85.47	85.08 8	3.07	88.12	85.52
CMR	81.55	81.80	81.20	81.46	80.30	80.59	80.00	80.19
Apparel	78.85	80.97	75.70	78.19	79.00	80.93	76.10	78.41
Books	75.00	72.21	81.40	76.48	74.50	71.55	81.60	76.17
DVD	76.65	73.98	82.30	77.89	76.15	72.76	83.50	77.73
Health	76.20	77.54	73.80	75.47	76.00	77.80	72.90	75.07
Video	80.10	77.93	84.10	80.85	79.30	76.62	84.50	80.33
	Feature presence with model selection				***Feature frequency with model selection***			
LMR	85.72	84.37	87.71	86.00	85.58	83.92	88.03	85.92
CMR	86.50	86.87	86.00	86.43	82.00	79.63	86.00	82.69
Apparel	82.50	85.71	78.00	81.67	80.00	80.61	79.00	79.80
Books	77.50	73.91	85.00	79.07	74.50	70.25	85.00	76.92
DVD	78.00	77.45	79.00	78.22	77.00	74.11	83.00	78.30
Health	79.00	80.21	77.00	78.57	78.00	78.57	77.00	77.78
Video	82.50	80.37	86.00	83.09	79.50	78.64	81.00	79.80

where the *score{pos|neg|obj}* indicates {positivity | negativity | objectivity} score of synset i for term A respectively and n represents number of synsets of term A. This interpretation of scores led to the calculation of sentence polarity and then may be used to estimate polarity of the whole document. The proposed approach is applied to movie-related reviews collected from Amazon and the results are compared with some machine learning-based classifier results. The results indicate that both the positive and negative sentences can be classified better with the SWN based algorithm by achieving an accuracy of 66%. In [77] the authors performed feature- based opinion mining of tweets. The results showed that character-grams performed better than word n-grams for Twitters' short text messages. The proposed approach combined Random Forests, Support Vector Machines, and Gradient Boosting Trees that were trained on a wide range of lexical and semantic features. These features were extracted using n-grams, brown clustering and part of speech tagged 1-gram models using SWN. The components were combined and optimized to create a separate version of the system for each of the completed subtask. The results obtained by Gradient Boosting Trees provided promising results in performing feature filtering.

In [78] polarity of tweets is determined by three different approaches: SentiWordNet, *Happiness score* [79] and machine learning based classifier. To determine the polarity of the tweets, two strategies were introduced. In the first strategy maximum, SWN scores for all the terms (either positive or negative) in the tweet are calculated. Then in the second phase, a sum of these scores, called a polarity score, is compared with two fixed-threshold values; positive-threshold (0.5) and negative threshold (0.5). The results showed that the SWN approach performed better than *Happiness score* and machine-learning based approach. Similar work on tweets is also reported in [56] in which SWN is merged with Subjectivity Lexicon [59] to calculate the polarity estimation of PoS in the given tweet. The authors comprehend the word-level polarity scores with different combinations of parts of speech like an adjective, adverbs, nouns, verbs to compute overall sentence polarity. A maximum accuracy of 58.96% was obtained after combining word-level evidence for the parts of the speeches mentioned above. It is also concluded that adjectives are strong indicators of describing the sentiment in a sentence. A different approach is used in [80] for a similar task by applying a sequential tagging approach. The authors used heuristic rules in combination with learned token labels to perform sentence level classification. SWN is specifically used to extract polarity information of the terms,

$$H(Ys) = \begin{cases} POS, & m(Ys) > 0, \\ NEG, & m(Ys) < 0, \\ NEU, & \text{otherwise.} \end{cases} \tag{3}$$

$$m(Ys) = \lambda count(POS, Ys) - (1 - \lambda) count(NEG, Ys), \tag{4}$$

where the function *count(l, L)* gives the number of the occurrences of specific label l in the label set L. The value of seed is set to 0.5 assuming that the target sentences to be classified contain the opinionated text. This assumption provided a way to develop the training data from the multiple corpus used in [80]. Different orientations of lexical relations are used to compute the polarity strength and achieved accuracy rate of 80.00%. The comparison of commonly used ten sentiment lexicons is given in Table 8.

Majority of the works at sentence level have been done on subjective text types, for example web-blogs and customer reviews in which the authors typically express their opinion freely. But the situation is different in the case of news articles as the newspapers publish objective data most of the time. Accordingly, journalists often have to be neutral and have to avoid using positive or negative sentiments. To address these issues, the work reported in [81–84] focused on news articles to separate positive and negative sentiments by considering smaller and larger word windows around the target entities. The polarity values of the quotes are computed by the sum of values of the words identified in them. A positive score classifies the quote as positive and a negative score results in classifying the quote being negative. The proposed approach is evaluated by considering both the complete set of positive and negative quotes. Before performing annotation, it was assured that what is the source and target of the sentiments. Furthermore, they also observed that there are three different possible aspects in newspaper articles -*author, reader and text*- and these aspects should be considered individually while analyzing sentiment. The sentence level features are investigated further to acknowledge and identify the subjective sentences in the whole review [84]. Similar research work was reported in [55, 85, 86] in which the sentence level emotion tags are used to assign the polarity value of the whole document. This emotion tagger that can be utilized in an emotion-based information retrieval system. In [87], the authors determined the polarity estimation at the sentence level by distinguishing between prior and contextual polarity. Furthermore, the authors focused on understanding which features are important for this task.

In [88, 89], the authors proposed a model for emotion detection in suicide notes with early fifteen different types of emotions. For experimental purpose, 1319 suicide notes between 1950 and 2011 were collected. SVM classifier in combination with SWN was engaged for binary classification and achieved an accuracy of 91.2% using cross-validation. A noteworthy work is reported in [90] which proposed a model to identify and estimate sentiment polarity at the sentence level using a set of rules that help in deriving polarity pattern. In order to learn the typed-dependency patterns, a set of a global class of sequential rules were utilized. For evaluation purposes, the dataset of [91] was used. The results showed an average F-measure of 85.37% using an adjectival modifier that also outperformed different feature selection approaches. In [20] experimental results on automatic annotation of emotions in the text are discussed in detail. Furthermore, a comparative analysis of several knowledge-based and corpus-based methods is also given in a very efficient way. The authors tried to identify the methods that work better for the annotation of emotions and sentiment terms and concluded that SWN-based methods proved better among other lexicons for the said purpose.

In a very recent and notable work reported in [77], the authors performed the sentiment classification of tweets and SMS (examples of unstructured data) using both unsupervised and

Table 8 Comparison of ten sentiment lexicons [95]

Sentiment resource	Reference	Entry size	Sentiment value
AFINN	[96]	2477 entries	No category is given. Each word has an integer score ranging between –5 (negative) and 5 (positive)
General Inquirer	[23, 24]	11, 789 words	Pstv/Ngtv/Pleasure/Pain/EMOT/etc. No numerical score
SentiWordNet	[7] [19]	117, 659 Synsets /155,287 words	Positive/negative/objective categories are given. Each category has a real-valued score within 0 and 1
Opinion Lexicon	[18]	6786 Words	Positive/negative categories are given. No numerical scores are given
Micro-WNOp	[97]	1105 synsets /1960 words	Positive/negative/objective categories are given. Each category has a real-valued sentiment score within 0 and 1
WordNet-Affect	[8, 22]	2872 Synsets /4552 words	Synsets are first categorized into emotion/mood/trait/behavior/etc., Further categorized into positive/negative/ambiguous/neutral categories. No numerical scores are given.
SO-CAL	[44]	6306 Words	No category is given. Each word has an integer score ranging between –5 (negative) and 5 (positive)
Subjectivity Lexicon	[59]	8221 Words	Positive/negative/both/neutral categories are given. No numerical scores are given
SenticNet	[16]	15,143 concepts	Pleasantness/attention/sensitivity/aptitude/polarity categories are given. Each category has a real-valued sentiment score within –1 and 1
SentiSense	[98]	2190 Synsets /4404 words	Joy/sadness/love/hate/despair etc. 14 emotion categories are given. No numerical scores are given

Table 9 Data by topics [101]

Topic	#Pos	#Neu	#Neg	#Irrelevant	Search Term
Google	2480	4815	1616	5280	#Google
Apple	2218	6102	4070	1940	#Apple
Twitter	2537	6245	2112	6410	#Twitter
Microsoft	2250	4487	3203	5430	#Microsoft
Total	9485	20,649	11,001	19,060	60, 195

supervised algorithms. In order to identify the sentiment of tweets, the unsupervised approach is being used. Furthermore, the SWN lexicon is used for the classification of tweets and movie review datasets. Each tweet (or movie review) was analyzed carefully and assigned a score using SWN information and PoS tagging. Several experiments were performed on various data sets. Table 9 gives detail of publicly available data sets. Some machine learning algorithms such as Naive Bayes, Maximum Entropy and Support Vector Machines are applied to perform sentiment detection and achieved accuracy of 80.68%. For supervised classification, the authors combine uni-gram, bi-gram for SMS data and achieved an accuracy of almost 67%. A similar technique is also used in a recent work [13] for cross-domain polarity estimation using the Amazon cat6 dataset.

Classifying the reviews based on the polarity is a common task in OM. Various machine learning techniques (using both supervised and unsupervised approaches) are proposed for classifying reviews [48]. These approaches analyze the data-set and assign each word a *positive*/*negative* score based on the number of entries in SWN. These scores are then used to estimate the final sentiment score for the full sentence. The documents are then classified as *positive* if the majority of the words in the sentence scored positive or *negative* if most words in the sentence are negative or *neutral* otherwise.

In [92, 93], the authors are concerned with the extraction of sentiment-oriented information from the reviews on social sites like MySpace, Twitter, YouTube, Digg, RunnersWorld and BBCForums. The authors used SentiStrength to compute sentiment strength. The results indicate that SentiStrength performed better than baseline approaches in both supervised and unsupervised cases.

It is noteworthy that in SentiWordNet, 90% of the terms are categorized as objective. These objective words (or terms) do not provide useful information to the classification modules. This inefficiency guided the researchers to develop a hybrid method [94]. The model is a composite of sentiment lexicon and machine learning approach to improve sentiment classification of tweets. The polarity score of the objective words is estimated from SWN (Eqs. (5) and (6)) and out of vocabulary (OOV) words extracted from tweet corpora. This polarity score is then compiled to create an *add-on lexicon*. The authors interpolated *uni*-gram vectors in SVM with sentiment lexicons. The results of the proposed lexicon improved the accuracy of sentiment classifications as compared to the baseline *uni*-gram model.

$$SWNScore(wi) = SWNScorePOS(wi) - SWNScoreNEG(wi). \quad (5)$$

$$Score(wi) = \begin{cases} Score_{POS}(wi), & \text{if } Score_{POS}(wi) > Score_{NEG}(wi), \\ (-1) \times Score_{POS}(wi), & \text{if } Score_{POS}(wi) < Score_{NEG}(wi). \end{cases} \quad (6)$$

In EQ. (6) we have,

$$Score_{POS}(wi) = \frac{P(positive|wi)}{P(positive)},$$

$$Score_{NEG}(wi) = \frac{P(negative|wi)}{P(negative)},$$

$$P(positive|wi) = \frac{No.\ of\ wi\ in\ positive\ tweets}{No\ of\ wi\ in\ datasets},$$

$$P(negative|wi) = \frac{No.\ of\ wi\ in\ negative\ tweets}{No\ of\ wi\ in\ datasets},$$

$$P(positive) = \frac{No.\ of\ positive\ tweets}{No.\ of\ all\ tweets},$$

$$P(negative) = \frac{No.\ of\ negative\ tweets}{No.\ of\ all\ tweets}.$$

4.3 Document level processing

Most of the documents express opinions about some real-life objects, for example, reviews or market surveys having open-ended statements that are useful resources to mine the opinions. Furthermore, these documents not only represent a single point of view but they typically contain multiple opinions as well. However, many sentiment analysis services produce a single opinion score to represent the whole document and hide the best insights. This is a major drawback in many opinion mining approaches at the document level since the useful insights may be hidden that may prevent clients from drilling down to extract the information. The majority of the early work at the document level OM is restricted to collecting data from news articles and product reviews [5, 7, 17, 18]. Due to the emergence and popularity of online social networks; disparate types of data collections, e.g., blogs and tweets encouraged the researchers to work in this field [1, 5, 17, 21, 29, 45, 46, 94, 102]. In this section, we will highlight these approaches that identify opinionated documents and classify according to their polarities — positive, negative, neutral.

In [103], the authors evaluated the product and business-oriented short reviews extracted from three different domains. SWN is used to create the domain-dependent feature set from the reviews. It was observed that the reduced features set has better performance while using a multi-domain dataset. Furthermore, the experimental results indicate that adding the contemporary sentiment features improve sentiment classification performance significantly. One of the limitations is that this approach is not efficient in differentiating ambiguous terms showing positive and negative sense at the same time. To deal with such problems contextual polarity [104] at word level is estimated after modification to the prior polarity of the terms. The polarity of the text at the document level can be calculated by the document discourse structure. In linguistics, the term *discourse* is referred to as a unit of language longer than a single sentence. In [105] the authors proposed a framework for sentiment analysis named as PATHOS that used discourse structure to interpret the text. This interpretation is then used to classify the polarity of the text. The PATHOS framework is comprised of three phases. In the first two phases, the documents are classified as positive or negative based on Word Sense Disambiguation (WSB) performed by identifying Parts of Speech (PoS) and lemmas of all the words. Then each word is assigned a weight by applying a discourse parser SPADE [106]. In the third and final phase, the overall sentiment of the document is computed as the weighted average of individual word scores retrieved from SWN. The estimated score played a vital role in the polarity estimation of the documents. After evaluating data sets from [106, 107] the results of the proposed framework outperformed the baseline sentiment classification approach. The results are depicted in Table 10.

Similarly, a two-step approach is followed by in [108] for the task of OM in TREC blog [109]. In the first step, the relevant documents are selected from the TREC Blog Track for each topic. After retrieving the relevant documents, logistic regression is applied to estimate the probability of each document whether the target document has the opinionated text or not. In the second step, the retrieval score and estimated probability are utilized to rank the documents. The features used for experimental purposes consist of about 5000 adjectives selected from SWN. After evaluating 11, 677 relevant documents, the overall prediction accuracy on the test set was 61%. The similar task for automatic classification of the product and movie reviews using SWN, at the document level is discussed in detail in the work of [100, 110, 111].

Lexicon-based and machine learning techniques provided promising results in OM tasks at the document level. Subjectivity lexicons can harvest into some effective domain-independent methods by giving a set of opinionated words. In general, a machine learning classifier trained on the data of a particular domain failed to perform well on data of another domain. Machine learning methods, however, can provide better performance under some given specific conditions but it requires a huge amount of data to train the classifiers.

Hybrid meta-classifier is used in [86] to perform sentiment classification of Spanish MuchoCine (MC) 16 corpus. The cor-

Table 10 Experimental results in terms of precision, recall, and F-measure for all positioners without offset [105]

	Positive			Negative			Overall	
Positioner	Prec	Recall	F1	Prec	Recall	F1	Accu	Macro F1
Baseline	0.562	0.775	0.651	0.637	0.395	0.488	0.585	0.569
Simple	**0.581**	0.770	0.662	0.659	0.445	**0.531**	**0.608**	**0.597**
SPADE I(1,0)	0.559	**0.810**	0.661	0.655	0.360	0.465	0.585	0.563
SPADE II(0.5, 0.5)	0.570	0.795	0.661	0.661	0.498	0.498	0.598	0.581
SPADE Extended	0.570	**0.810**	**0.669**	**0.672**	0.390	0.494	0.600	0.582

pus is translated into MuchoCine English (MCE) corpus, which consists of 3878 movie related reviews collected from the MuchoCine website. For both the corpora, SVM and NB classifiers are applied for supervised learning. For unsupervised learning, SWN based classifier is used to classify the MCE movie reviews. Finally, both approaches were aggregated to build a meta-classifier. The standard schema for document weighting like *tf-idf*, Term Occurrence (TO), Term Frequency (TF), and Binary Occurrence (BO) are used as decision parameters. Experimental results showed that the SVM classifier outperformed NB for both corpora. The *tf-idf* measure is identified better among the parameters. The best precision of 0.8773 and 0.8776 on MC and MCE respectively is obtained by SVM classifier using *tf-idf* feature.

The work reported in [141] determined the sentiment strength of tweets expressing an opinion on popular events. The authors observed the tweets on top 30 general events and concluded that the increased negative sentiment associated with the text makes the events more popular than the others. For evaluation purpose, the authors utilized a part of spinn3r[1)] dataset containing 34, 770, 740 English language tweets collected from 2, 749, 810 different accounts. SWN based SentiStrength dictionary [92, 93] is exploited to perform Sentiment classification. For each topic, the authors computed sentiment strength for all kinds of tweets regardless of the fact the tweets having a positive or negative sentiment. The results led to different conclusions regarding the difference in average sentiment strength among different events.

In [112], the authors created a new senti-lexicon[2)] for restaurant reviews. This lexicon comprised of intensifiers, *uni*-gram, *bi*-gram, and negations. A group of improved NB classifiers was applied to leverage the accuracy gap in the classification of positive and negative terms. A total of 70, 000 of review documents are collected from different restaurant sites to perform sentiment classification. The proposed approach outperformed baseline methods and achieved an accuracy of 81.2%. A different methodology comprised of traditional NLP-techniques and Semantic Web technologies is proposed in [113] for feature-based OM at the document level. The proposed model works in three stages: (i) identification of sentiment related features using a knowledge-based repository e.g. ontology (ii) assigning the polarity to each feature based on its position in each user's opinion using SWN and (iii) extracting the opinions using a novel approach based on vector analysis. To identify the correct features, a populated ontology is used. The results outperformed other baseline approaches and achieve higher accuracy.

Web 2.0 is regarded as the revolutionary development of the Internet. This development includes knowledge building by actively involving the learners and users, switching of static web pages to dynamic, the introduction of social media platforms, etc. Social media provides a platform that allows users to share their ideas and suggestions and interact with other users worldwide in a virtual community. This media provides a quite different perspective in comparison with conventional websites where the users are limited only to the passive viewing of content [103, 109, 114, 115]. These platforms provide the opportunity to millions of users to share their ideas on some event or some product. As a result, a huge amount of user-generated content is developed. Emotions seem to frequently appear in these messages to express a variety of sentiment that has now become a rich and useful resource to perform OM tasks [4, 11, 16, 20]. Efficient algorithms are developed to extract general sentiment and its strength from this informal communication to understand the role of emotion mining. In this subsection, we will discuss some of the notable works in detecting the sentiment strength. Much of the recent research in OM has focused on finding low-level features that help in predicting the polarity of phrase, sentence or text. To identify and detect inappropriate and anomalous effective words, in [116, 117] the authors proposed a model to detect radical content. The results showed that SWN can be used for analyzing sentences having radical content appearing on web.

An efficient hybrid approach of lexicon and machine-learning technique is proposed in [100, 118] to classify paragraphs of movie and product reviews into descriptive or comment categories. The taxonomy used in combination with SWN to evaluate the polarity of the words in the reviews. It was concluded that *average on sentence* and *average on review* methods yield an improved result over the term count method [119]. In [120], SWN is exploited in some interesting tasks of mood classification in the lyrics dataset. For evaluation purposes, 185 lyrics are collected, each belonging to one of the four different classes. Both the average and ratio of the positive and negative scores of the representative words in lyrics were considered as the sentiment features. The results achieved an accuracy rate of 78% using NB, KNN and SVM classifiers using the features mentioned above. The results are shown in Fig. 5. Similarly, in [84, 121], the authors performed a series of experiments in a regression framework to compute prior polarities of a word using its posterior. It was concluded that a weighted average is the best strategy to be followed. In genre-based classification, the use of sentiment features enables us to obtain good classification accuracy with significantly fewer features. The use of SWN for sentiment feature extraction of documents is considered as an efficient pre-processing stage for genre-based classification.

The authors in [21, 86, 120] used a dataset of [122] and filled the gap between the opinion identification and users' behavior detection about some product. An innovative algorithm called SentiStrength is developed using SWN to compute the sentiment strength from the informal English text. Furthermore, it also analyzes the grammar and spelling styles from cyberspace. This algorithm predicted both positive and negative emotions with 60.6% and 72.8% accuracy respectively when applied to 1000 comments extracted from MySpace[3)]. The results obtained by SentiStrength are compared with various algorithms and conclude that the proposed algorithm performed better than the wide range of general machine learning approaches. The results are shown in Table 11.

In a recent work reported in [113], stock prices are predicted using SWN by analyzing financial reviews as target entities.

[1)] Web and Blog datasets https://snap.stanford.edu/data/other.html
[2)] Big Data Lab http://irlab.sejong.ac.kr/res-sentiwordnet
[3)] Featured Content on Myspace (last accessed on July 4, 2017) https://myspace.com

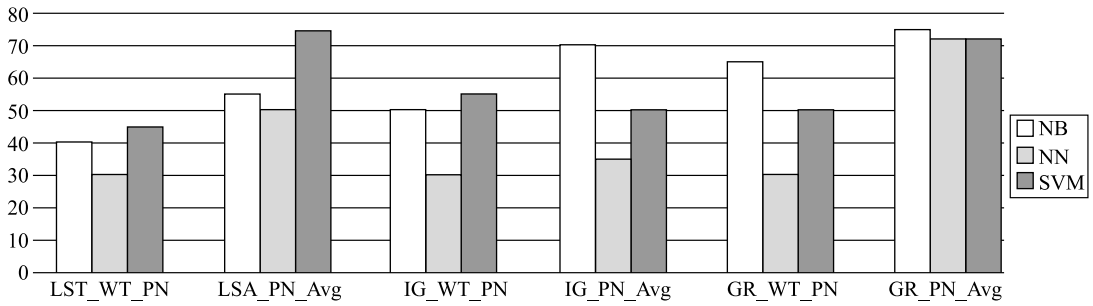

Fig. 5 Accuracy of Naïve Bayes classifier (NB), *k*-nearest neighbor (KNN), and support vector machine (SVM) classifiers using 10% features [120]

Table 11 Performance of various classification algorithms on positive sentiment strength detection [118]

Algorithm	Optimal features	Accuracy /%	Accuracy 1 class/%	Corr	Mean error
SentiStrength (Standard Config. 30 runs)	-	60.6	96.9	0.599	22.00
Simple logic regression	700	58.5	96.1	0.577	23.2
SVM (SMO)	800	57.6	95.4	0.538	24.4
J48 Classifier	700	55.2	95.9	0.548	24.7
jRIP rule-based classifier	700	54.3	96.4	0.476	28.2
SVM regression (SMO)	100	54.1	97.3	0.469	28.2
AdaBoost	100	53.3	97.5	0.464	28.5
Decision Table	200	53.3	96.7	0.431	28.2
Multilayer Perceptron	100	50.0	94.1	0.422	30.2
Naive Bayes	100	49.1	91.4	0.567	27.5
Baseline	-	47.3	94.0	-	31.2
Random	-	19.8	56.9	0.016	82.5

The respective reviews are collected from financial social networks like StockTwits[4)] and SeekingAlpha[5)]. The authors used the rule-based model, VADER for sentiment classification and the results showed that VADER not only outperformed machine learning methods but it is faster also.

To perform the task of *lexicon creation*, a list of words (seed words) is used. This list is extended using synonyms and antonyms of the seed words [2, 27, 32] extracted from WordNet [8, 22] dictionary. The extension of the list is an iterative process and terminated when there are no more antonyms (or synonyms) in the dictionary. SWN is also used for the same purpose in the notable work in [44, 105, 123]. Similar work is also observed in [66, 117] in which the authors presented some novel approaches for the dynamic creation of sentiment word lists. In these approaches, the seed words are first mapped into a continuous latent space. This mapping then serves as input to a multilayer perceptron (MLP) classifier that is trained using sentiment-annotated words. Experimental results using manually annotated EmoLex corpus [124] showed that these approaches compare favorably with SWN.

4.4 Aspect and clause level processing

In aspect level OM, the objective is to detect and aggregate sentiments and opinions on features (or aspects) of the target objects mentioned in target documents. Hence, preliminary initiatives were undertaken to identify the hierarchical relationship among the **target** and its corresponding aspects. OM at this level deals with the aspects (or features) of the given entity in reviews and classifies the reviews based on features. There are some parts of speech like *adjective*, *adverb*, *verb*, and *noun* that may be considered in finding the relative opinion in reviews. These parts of speech play important role in expressing the *opinion* or *sentiment* in the text [18, 27].

Domain-specific resources contain a set of aspects and features such as dependency patterns, taxonomy, and cues of features. In [126], the authors proposed a novel approach for product feature extraction using unsupervised techniques in feature-oriented opinion reviews. For each domain review corpus, the authors coined a novel term *Domain Entity* (DE) to represent the specific domain. The similarity between the feature and domain entity plays an important role in creating an association between a feature and its associated domain. The experimental results showed that the proposed approach outperformed the state-of-the-art methods in performing feature extraction. They also introduced another novel term *similarity measure* (PMI–TFIDF) that depicts the degree of association of corresponding features and domain terms. The results also showed that PMI–TFIDF presented better distinction ability as compared with conventional PMI measures. Similarly, in [127] the authors extract the opinions with a joint aspect/sentiment (JAS) model. Latent Dirichlet allocation (LDA) is combined with a sentiment before performing topic extraction and subjectivity classification. For evaluation purposes, some experiments were performed on restaurant and hotel reviews using MPQA and SWN. The output of the proposed model superseded with other models such as the maximum entropy (MaxEnt)-LDA, Aspect, and Sentiment Unification Model(ASUM).

It is observed that the important aspects of a product are those that are frequently commented on by consumers. Keeping in view the above observation, in [112] the authors developed an aspect ranking algorithm. The algorithm considers the aspect frequency (*af*) that helps in finding the important aspects. The authors also performed several experiments on 11 popular products of four different domains. The proposed approach gives prominent results in the identification of important aspects. In [113] the authors proposed a different methodology for feature extraction and aspect-based OM for movie reviews, using movie ontology. Each of the movie-related features is as-

[4)] StockTwits. Never Trade Alone (last time accessed on July 4, 2017) https://stocktwits.com
[5)] Stock Market Insights | Seeking Alpha (last accessed on July 4, 2017) https://seekingalpha.com.

signed a relative sentiment score according to its position in the review i.e. beginning, middle, or end. SWN is used to compute the overall sentiment score of the review. Experimental results were performed on movie datasets from [3, 91] and achieved a classification accuracy of 89.6%.

A similar work reported in [91, 128] that focused on ontology-based sentiment aggregation using four-level hierarchical relationships. Aspects (or features) appearing at a higher level were assigned high weights and those occurring at lower levels were assigned lower weights. ConceptNet 5 ontology is used to construct the required ontology. SWN is utilized in deciding the subjectivity of a word. A compiled dataset was developed from 584, 986, and 1000 user reviews on the automobile, digital camera, and general purpose software respectively. The domain ontology was mapped with WordNet for calculating similarity measures. The overall review polarity is then computed using the bottom-up approach by combining the node polarities. Then the authors performed experiments on datasets from [3, 91] and report an improved accuracy of 7.55% over SVM classifier and outperforming other 11 approaches.

In [129] authors proposed a linguistic model for sentiment analysis of messages posted on various online discussion forums. These messages consist of independent clauses having opinionated text describing different features of a target object. To analyze these messages, the authors proposed the clause-level sentiment algorithm given in Fig. 6. The algorithm generates a dependency tree by splitting each sentence into clauses. Then the contextual sentiment score is determined for each clause by exploring grammatical dependencies in the tree. In the end, prior sentiment scores are calculated for the words derived from SWN. Negative sentiment in a sentence is also handled in a very efficient way in this model, for example, the term *not superb* is assigned a lower negative sentiment score than the term *not good*. The authors performed experiments on a dataset of movie reviews and achieved an accuracy of 81%.

Sentiment analysis on review documents needs to focus multiple sentiments towards different aspects of the reviewed entity at the clause level. For example in the work of [28] three approaches namely Enhanced Emoticon Classifier (EEC), SentiWordNet based classifier (SWNC) and Improved Polarity Classifier (IPC) are applied successively to categorize the tweets. Initially, the EEC was used on a collection of 145 emoticons (70 positive, 75 negative) to perform classification. Then IPC is applied on the tweets categorized using Opinion Lexicon[6] and Bill McDonald dictionaries [130]. Then the SWN dictionary is used to classify the neutral tweets by IPC. For evaluation purpose, a total of 2116 tweets on six personalities/nations was collected to create a total of six datasets namely, Nawaz Sharif, Imran Khan, Tom Cruise, Mahendra Singh Dhoni, Pakistan, and America. The result showed that SWN based method achieved an average high accuracy of 85.6% on all six datasets among the other techniques.

```
FOR each sentence
    Perform Semantic Annotation
    Assign prior sentiment scores to words
    Generate grammatical dependencies
    Break a sentence into clauses
    FOR each clause
        Calculate clause level Sentiment Score (Single Aspect)
            Subject or Object (Adjective+Noun)
            Verb phrase (Adverb+Verb)
            Predicate (Verb phrase+Object/complement)
            Clause (Subject and Predicate)
        Determine Aspenct
    END FOR
END FOR
```

Fig. 6 Algorithm for clause level semantic analysis [129]

In [131, 132], a novel based approach is proposed for OM using SWN. The authors arranged the aspects of different reviews in the tourism domain as very few researchers focused on this field. The hotel reviews are collected from an American based travel and restaurant website TripAdvisor. To derive the most appropriate aspects of the hotel reviews an efficient algorithm is designed. Each review is carefully analyzed to extract the correct features and opinions having aspects that are assigned an SWN based score. Finally, for each aspect, the cumulative score is computed and then the results are visualized by a graph.

A comprehensive overview is given in [112] on various aspect extraction techniques and approaches used in OM. These techniques are categorized based on the adopted approaches. The review encompasses not only the performances of different techniques but also allow the reader to compare the accuracy with other state-of-the-art approaches.

4.5 Concept level processing

Concept-based approaches in OM deal with opinion analysis of text using knowledge resources like web ontologies or semantic networks. These resources help in retrieving the affective information associated with the opinions. Unlike conventional OM and syntactical techniques, concept-level OM deals in detecting the sentiments specifically expressed in the delicate and tenuous way, e.g., detecting the sentiments not expressing the emotions clearly but are implicitly linked to other concepts [16, 133]. In this section, we discuss some of the notable work at the concept level.

In [134] the authors proposed a multi-layer methodology to analyze intra-day directional transaction of currency-pair to and from foreign exchange (FOREX) markets. SWN is used here to obtain sentiment scores along with WordNet to extract hypernyms *tf-idf* to leverage the term. An SVM based classifier is applied to perform sentiment classification of news headlines. A corpus of financial news from MarketWatch[7] and foreign exchange historical data in the form of Euro/ US dollars (USD) is collected to conduct experiments. The results (see Fig. 7) indicate that the SVM classifier outperformed NB, and k-nearest neighbor (KNN) classifiers with an accuracy of 83.33%.

Similarly in [124] an approach for early detection of emerging political topics using Twitter trends. Sentiment lexicon like

[6] Opinion Mining, Sentiment Analysis, and Opinion Spam Detection (last time accessed on July 16, 2017) https://www.cs.uic.edu/~liub/FBS/sentiment-analysis.html

[7] MarketWatch: Stock Market News - Financial News(last accessed on September 3, 2017) MarketWatch.com

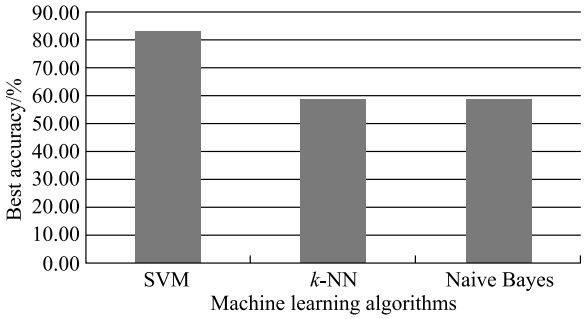

Fig. 7 Accuracy levels for different machine learning algorithms [89]

SenticNet3[8] and SWN were used to compute the polarity of tweets. To compute the polarity of new and novel words, a relational graph is plotted using data of political tweets posted at different periods. For experimental purpose, about 4,000,000 tweets were collected during the Germany parliamentary election with a concentration on nine political parties. It is concluded that the proposed approach explored the emerging patterns in controversial political issues quite efficiently.

Table 12 depicts the distribution of reviewed articles with respect to published year wise and in Table 13 gives detail of reviewed articles based on different tasks and sub-tasks using SWN as a lexical resource. Table 14 shows the main classification technique used by various articles over the years. As can be seen, the SVM classifier is the most popular choice followed by Naïve Bayes and dictionary-based approaches.

5 Discussion

Gradually, OM research has distinguished itself as a separate disciplined field. This field mainly focuses on semantic orientations and affective/emotional aspects associated with natural language that is the reason it merely depends on a profound understanding of the text. Moreover, it also involves analysis of NLP in several different dimensions like content-, concept- and context-based with support of lexical resources to produce time-efficient results.

In this paper, we have summarized many interesting and challenging tasks in the field of OM in which SWN is used as a lexical resource. The article is organized based on tasks at different granular levels of OM as shown in Table 5. It is obvious from the details given in Table 5 that concept level tasks

Table 12 Distribution of articles published year wise

Year	References	No. of articles
2004	[8, 122, 136]	3
2005	[6, 58, 59]	3
2006	[7, 108]	2
2007	[67, 87, 137]	3
2008	[7, 12, 20, 35, 36, 50, 80, 94, 137]	9
2009	[17, 45, 52, 72, 82, 107, 109, 119, 135]	9
2010	[4, 19, 48, 62, 93, 110]	6
2011	[9, 26, 31, 44, 71, 79, 83, 92, 104, 105, 111, 118]	12
2012	[1, 2, 16, 31, 37, 46, 60, 63, 65, 78, 84, 85, 90, 92, 99, 112, 116, 120]	18
2013	[5, 21, 29, 32, 40, 43, 46, 51, 53, 64, 66, 82, 83, 86, 110, 126]	23
	[16, 84, 91, 116, 120, 133, 138]	
2014	[16, 28, 33–35, 49, 92, 94, 100, 103, 128–130, 139]	14
2015	[11, 18, 54, 89]	4
2016	[28, 53, 55, 90, 115, 116, 127, 130, 139, 140]	10
2017	[70]	1
2018	[13, 69, 101]	3
Total		**121**

Table 13 Distribution of research articles (reviewed in this article) with respect to tasks/subtasks in which SentiWordNet (SWN) used as sentiment lexicon (SL) for opinion mining (OM)

Tasks or subtasks	References	No. of articles
Polarity detection/Sentiment classification	[60, 63, 69, 90, 105]	5
Opinion classification, disambiguation & extraction	[21, 28, 30, 49, 52, 55, 62, 76, 80, 94]	19
	[77, 100, 103, 112, 113, 118, 119, 131, 135]	
Lexicon creation	[71, 104, 126]	3
Cross-domain OM	[66, 83]	2
Subjectivity classification	[16, 20, 55, 82, 88]	5
Source classification	[64, 82]	2
Feature extraction	[48, 72, 103, 110, 129]	5
Multi-lingual OM	[76, 85, 86]	3
Aspect and entity extraction	[113]	1
Sentiment Strength	[70, 92, 103, 114, 118]	5
Applications of OM (SWN based)	[16, 44, 92, 93, 100, 104, 118, 120]	18
	[84, 116, 117, 121, 122, 132, 139, 142–144]	

[8] Details are in (last accessed on September 23, 2017) http://sentic.net/senticnet-3.0.zip

Table 14 Distribution of articles based on classifier used

Classification technique used	References	Number of papers
Support vector machine (SVM)	[21, 28, 39, 44, 48, 52, 56, 60, 62–65, 70, 94]	36
	[30, 77, 78, 80, 83, 85, 86, 88, 92, 93, 100, 101, 105]	
	[16, 84, 89, 103, 104, 112, 119, 129, 132]	
Naïve Bayes (NB)	[16, 21, 28, 48, 52, 63, 69, 86, 92, 93, 100, 101, 112, 114]	14
Dictionary based approaches (DBA)	[28, 44, 64, 66, 92, 125, 126, 131]	8
Corpus based	[20, 35, 69, 72, 82, 108, 116]	7
Logistic regression	[66, 83, 84, 92, 93]	5
SVM-SMO	[39, 70, 92, 93, 127]	5
Semantic relationship	[71, 110, 116, 132]	4
Rule based classifier & random forests	[67, 77, 94, 130]	4
Semantic orientation	[48, 62, 80]	3
Maximum entropy	[56, 63, 101]	3
TF_IDF & term counting	[89, 103, 118]	3
K-nearest neighbor (KNN)	[52, 138]	2
Ontology	[91, 113]	2
SentiStrength	[92, 141]	2
Dependency tree	[135]	1
Linear discriminant analysis (LDA)	[66]	1
Class sequential rules	[90]	1
Kernel methods	[129]	1

have been addressed in very few studies. It is observed that the researchers show more interest in the task of opinion classification and extraction than polarity detection, as shown researchers show more interest in the task of opinion classification and extraction than polarity detection, as shown in Table 13. It is also concluded that among various tasks (and sub-tasks) of OM, cross-domain analysis [66, 83] and source classification [65, 82] need more effort from the researchers and analysts.

Among the classifiers used in OM tasks, SVM is the most widely used classifier. This is because SVM avoids overfitting in the results and it also proved itself efficient in solving high dimensional data problems as compared to other techniques.

Besides the role of SWN discussed in this review paper, currently, SWN is involved in very interesting applications like sarcasm detection [140, 145] and human-agent interaction [130, 146], extracting symbolic meaning from gesture [133], discovering hidden knowledge from the text, etc. However, there is still a lot of work to be done in using SWN at global and business levels. The global level includes spam and rumor detection, big data opinion analysis in cloud computing while business level includes the decision making for candidates based on endorsement given by former employers and preparation of appraisal reports for an employee based on a customer-centric approach.

While discussing OM in general, we highlighted some of the future work in the field of OM based on SWN as a lexical resource. First, while OM focused on textual data, future data will predominantly be stored in much more complex data types and opinion mining will have to cope with this increasing volume of structured data. Furthermore, an emerging aspect is "patterns in data" that has become an important area for the researchers. Second, it is also observed that the data to be analyzed is extracted from several sources. To solve this issue, an emerging trend is to introduce efficient data preprocessing and integration approaches to certifying that the "patterns in data" found are "legal" on the complete set of data objects and not just on a particular subset. The users will be able to adjust the available options and knobs to get efficient results if future OM methods succeed in knobbing and resolving the issues mentioned above. Furthermore, the major objective is to achieve a user-friendly OM system with minimum parameters that will result in enhancing the usability by finding new types of patterns.

We believe that there several interesting new challenges in OM since no human being can foretell the future. We have tried our best to cover new avenues in OM, in general.

• Dominant and open challenges

In this section, several issues and challenges related to OM using SWN as a lexical resource are discussed in detail that was found in the articles surveyed in this paper.

1) There is so much noise in the data collected from various resources, like wrongly spelled terms and unstructured format of the words. Unfortunately, there is no publicly available system for noise removal and automatic spelling correction. This noise makes the preprocessing step to consume maximum time to convert this unstructured raw data into a manageable format.
2) In order to extract the precise features of the data, most of the researchers have not utilized the optimization techniques. Therefore, a combination of various machine learning models and optimization techniques can be used to extract characteristic features.
3) There are very few attempts in developing OM systems for the non-English domain. Most of the researchers have carried out their work in the English language. Therefore, these resources should be portrayed in a language other than English to perform cross-lingual OM.
4) One of the main challenges is the validation of the proposed methods while performing OM tasks related to e-marketing. The researcher should be given access to all

the relevant information about some products to validate the proposed method. This kind of information surely helps in finding an association between sales of the product and its helpfulness.

6 Conclusion

In this paper, the role of SWN is analyzed in different tasks related to opinion mining (OM) in detail. This review is categorized in OM tasks based on word, sentence, document, aspect, and concept-based levels while analyzing the multi-purpose OM tasks. Lexicon-based approaches are more popular among researchers since these approaches require no training data and are more suitable to a wider range of related domains than the conventional supervised approaches. However, lexicon-based approaches used in OM tasks have two limitations. First, the lexicons have a finite number of words that may create a problem while extracting sentiment from very dynamic environments such as the blogosphere. Secondly, sentiment lexicons assign a fixed sentiment score to the words without taking into account its context in the sentence. The words may express different sentiments in a different context. For example, the word *great* may have **negative** sentiment while talking about some *problem* while it have quite **positive** affect in the context of *justice*.

It is also concluded that SWN has proved itself a reliable resource for sentiment strength detection in blog postings, product reviews, short texts, and social web, without using the training model. SWN also helps in creating a dynamic representation of the context of the words by making use of the pre-assigned score and polarity of words in the lexicon. The approaches and the procedures discussed in the paper are limited to the major tasks of OM by exploiting only one lexical resource SWN. There is a need for a comprehensive review of the role of other lexical resources in OM to have a comparative study.

One of the main contributions of the paper is to present a list of available public datasets for OM that makes this review different and unique from other reviews of the same kind. This comprehensive survey will be quite obliging to novice researchers to address challenging issues conclusively. Furthermore, it will also help in finding the most common challenges to propose a new solution.

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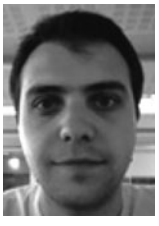

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